

IV Semester			
S. No	Course Code	Course Title	Credit Hours
1.	SSAC-221	Soil, Plant and Water Analysis*	2(0+2)
2.	SSAC-222	Problematic soils and their Management	2(1+1)
3.	AECON-221	Principles of Agricultural Economics and Farm Management	2(2+0)
4.	AGRON-221	Crop Production Technology–II (<i>Rabi</i> crops)	3(1+2)
5.	AGRON-222	Water Management	2(1+1)
6.	AENGG-221	Farm Machinery and Power	2(1+1)
7.	AGEXT-221	Entrepreneurship Development and Business Communication	3(2+1)
8.	GPB-221	Basics of Plant Breeding	3(2+1)
9.	HORT-221	Production Technology for Vegetables and Spices	2(1+1)
10.	NCC-IV/ NSS-IV PHED-IV	NSS/NCC/Physical Education & Yoga Practices**	1(0+1)
Total Credit			21(11+10)+1**
*Skill Enhancement Courses; **NC: Non-gradual courses			

SSAC-221	Soil, Plant and Water Analysis*	2(0+2)
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Practical

General considerations of analytical determination. Soil sampling and sample preparations for analysis. Basic principles of the instruments used in soil, plant and water analysis. Determination of soil pH, electrical conductivity (EC) Organic Carbon and CEC of soil. Determine available nitrogen, phosphorus, potassium, sulphur, micronutrients in soil. Soil testing and interpretation - The basics. Plant sampling and sample preparation for analysis. Rapid plant tissue test for N, P and K. Estimation of total nitrogen, phosphorus, potassium, in plants. Chemical analysis of irrigation water, Parameters for water quality assessment, water quality guidelines for irrigation.

S.No.	Topic	No. of lectures
1.	General considerations of analytical determination	3
2.	Soil sampling and sample preparations for analysis	2
3.	Basic principles of the instruments used in soil, plant and water analysis	3
4.	Determination of soil pH and electrical conductivity (EC) of soil	1
5.	Determination of organic carbon in soil sample	1

6.	Determine CEC of soil by using Neutral Normal Ammonium Acetate	2
7.	Determine available nitrogen in soil	1
8.	Determination of available phosphorus in soil	1
9.	Determination of available (Exchangeable) potassium in soil	1
10.	Determination of available sulphur in soil	1
11.	Determination of available (DTPA Extractable) Zinc, Copper, Iron and Manganese	1
12.	Soil testing and interpretation - The basics	1
13.	Plant sampling and sample preparation for analysis	2
14.	Rapid plant tissue test for N, P and K	2
15.	Estimation of total nitrogen in plant	1
16.	Determination of total phosphorus in plant	1
17.	Estimation of total potassium in plant	1
18.	Chemical analysis of irrigation water	3
19.	Parameters for water quality assessment	2
20.	Water quality guidelines for irrigation	2
	Total practical classes	32

SSAC-222	Problematic soils and their Management	2(1+1)
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Objectives

1. To acquaint the students about various problem soils like degraded soils, acid soils, saline soils, alkali soils, eroded soils, submerged soils, polluted soils. Also to impart knowledge about remote sensing, GIS, Multipurpose tree and Land capability classification
2. To give hands on training about estimation of various soil and water quality parameters associated with problem soils.

Theory

Soil quality and health, Distribution of Waste land and problem soils in India, Categorization of Problem soils based on properties. Reclamation and management of Acid soils, Saline, Sodic soils, Acid Sulphate soils, Eroded and Compacted soils, polluted soils. Contaminated soils (Pesticide contamination, Heavy metal contamination), Mined soils (Coal mined, Oil mined), Management of Riverine soils, Waterlogged soils, Irrigation water – quality and standards, utilization of saline water in agriculture. Use of Remote sensing and GIS in diagnosis and management of problem soils. Irrigation and water quality. Multipurpose tree (MPT) species, bio remediation through MPTs of soils, land capability and classification, land suitability classification.

S.No.	Topic	No. of lectures
1.	Soil quality and health, Distribution of Waste land and problem soils in India	1
2.	Categorization of Problem soils based on properties.	2
3.	Reclamation and management of Acid soils, Saline, Sodic soils, Acid Sulphate soils, Eroded and Compacted soils, polluted soils	3
4.	Contaminated soils (Pesticide contamination, Heavy metal contamination), Mined soils (Coal mined, Oil mined),	1
5.	Management of Riverine soils, Waterlogged soils,	1
6.	Irrigation water – quality and standards utilization of saline water in agriculture.	2
7.	Use of Remote sensing and GIS in diagnosis and management of problem soils	2
8.	Irrigation and water quality	1
9.	Multipurpose tree (MPT) species, bio remediation through MPTs of soils	1
10.	land capability and classification	1
11.	land suitability classification	1
	Total theory classes	16

Practical

Determination of pHs and EC of saturation extract of problematic soil. Determination of redox potential in soil, Estimation of water soluble and exchangeable cations in soil and computation of SAR and ESP and characterization of problematic soil. Determination of Gypsum requirement of alkali / sodic soil. Determination of lime requirement of acidic soil. Determination of Quality of irrigation water (pH, EC, Ca, Mg, Na, CO₃, HCO₃, Cl, SAR and RSC), Determination of nitrate (NO₃⁻) from irrigation water, Determination of dissolved oxygen and free carbon dioxide levels in water samples.

Practical

S.No.	Topic	No. of lectures
1.	Determination of pHs and EC of saturation extract of problematic soil	2
2.	Determination of redox potential in soil	1
3.	Estimation of water soluble and exchangeable cations in soil and computation of SAR and ESP and characterization of problematic soil.	3
4.	Determination of Gypsum requirement of alkali / sodic soil.	1
5.	Determination of lime requirement of acidic soil	1

6.	Determination of Quality of irrigation water (pH, EC, Ca, Mg, Na, CO ₃ , HCO ₃ , Cl, SAR and RSC),	5
7.	Determination of nitrate (NO ₃ ⁻) from irrigation water	1
8.	Determination of dissolved oxygen and free carbon dioxide levels in water samples.	2
	Total practical classes	16

Suggested readings

1. Agarwal, R.R., Yadav, J.S.P. and Gupta, R.N. (1982). Saline Alkali soils of India, ICAR, AGROBIOS (India).
2. Brady Nyle C and Ray R Well., 2014. Nature and properties of soils. Pearson Education Inc., New Delhi.
3. Cirsan J. Paul., 1985,. Principles of Remote Sensing. Longman, New York
4. Indian Society of Soil Science., 2002. Fundamentals of Soil Science. IARI, New Delhi.
5. Osman, Khan Towhid., 2018., Management of Soil Problems. Springer publication
6. Srivastava, V. C., 2002. Management of Problem Soils -Principles and Practices New Delhi

AECON-221	Principles of Agricultural Economics and Farm Management	2(2+0)
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Objective

1. To aware the students about broad areas covered under agricultural Economics and farm management
2. To impart knowledge on judicious use of resources for optimum production

Theory

Economics: Meaning, scope and subject matter, definitions, activities, approaches to economic analysis; micro- and macro-economics, positive and normative analysis. Nature of economic theory; rationality assumption, concept of equilibrium, economic laws as generalization of human behavior. Basic concepts: Goods and services, desire, want, demand, utility, cost and price, wealth, capital, income and welfare. Agricultural economics: meaning, definition, characteristics of agriculture, importance and its role in economic development. Agricultural planning and development in the country. Demand: meaning, law of demand, demand schedule and demand curve, determinants, utility theory; law of diminishing marginal utility, equi-marginal utility principle. Consumer's equilibrium and derivation of demand curve, concept of consumer surplus. Elasticity of demand: concept and measurement of price elasticity, income elasticity and cross elasticity. Production: process, creation of utility, factors of production, input output relationship. Laws of returns: Law of variable proportions and law of returns to scale. Cost: Cost concepts, short run and long run cost curves. Supply: Stock v/s supply, law of supply, supply schedule, supply curve, determinants of supply, elasticity of supply. Distribution theory: meaning, factor market and pricing of factors of production. Concepts of rent, wage, interest and profit. National income: Meaning and importance, circular flow, concepts of national income

accounting and approaches to measurement, difficulties in measurement.. Money: Barter system of exchange and its problems, evolution, meaning and functions of money, classification of money, money supply, general price index, inflation and deflation. Economic systems: Concepts of economy and its functions, important features of capitalistic, socialistic and mixed economies, elements of economic planning. Forms of business organizations, international trade and balance of payments. GST and its implication on Indian economy.

Lecture Schedule: Theory

S. No.	Name of Topic	No. of Lectures
1.	Economics: Meaning, scope and subject matter, definitions, activities, approaches to economic analysis; micro- and macro-economics, positive and normative analysis.	2
2.	Nature of economic theory; rationality assumption, concept of equilibrium, economic laws as generalization of human behavior.	1
3.	Basic concepts: Goods and services, desire, want, demand, utility, cost and price, wealth, capital, income and welfare.	2
4.	Agricultural economics: meaning, definition, characteristics of agriculture, importance and its role in economic development.	2
5.	Agricultural planning and development in the country.	2
6.	Demand: meaning, law of demand, demand schedule and demand curve, determinants, utility theory; law of diminishing marginal utility, equi-marginal utility principle.	2
7.	Consumer's equilibrium and derivation of demand curve, concept of consumer surplus.	1
8.	Elasticity of demand: concept and measurement of price elasticity, income elasticity and cross elasticity.	2
9.	Production: process, creation of utility, factors of production, input output relationship.	2
10.	Farm Management- Meaning, Definition, objective, its relation with other sciences. Concept and Types of Production functions	1
11.	Laws of returns: Law of variable proportions and law of returns to scale.	1
12.	Cost: Cost concepts, short run and long run cost curves.	2
13.	Supply: Stock v/s supply, law of supply, supply schedule, supply curve, determinants of supply, elasticity of supply.	2

14.	Distribution theory: meaning, factor market and pricing of factors of production. Concepts of rent, wage, interest and profit.	2
15.	National income: Meaning and importance, circular flow, concepts of national income accounting and Approaches to measurement, difficulties in measurement.	2
16.	Money: Barter system of exchange and its problems, evolution, meaning and functions of money, classification of money, money supply, general price index, inflation and deflation.	2
17.	Economic systems: Concepts of economy and its functions, important features of capitalistic, socialistic and mixed economies, elements of economic planning.	2
18.	Forms of business organizations, international trade and balance of payments. GST and its implication on Indian economy.	2

Suggested Readings

1. Johl, S. S. and T. R Kapur. 2009. Fundamentals of Farm Business Management. Kalyani Publishers
2. S. Subha Reddy, P. Raghu Ram, T. V. Neelakanta and I. Bhvani Devi. 2004. Agricultural Economics. Oxford & IBH publishing Co. Pvt. Ltd

AGRON-221	Crop Production Technology–II (<i>Rabi</i> crops)	3(1+2)
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Objectives

1. To impart basic and fundamental knowledge on principles and practices of rabi crop production.
2. To impart knowledge and skill on scientific crop production and management

Theory

Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield of *Rabi* crops. Cereals- wheat and barley. Pulses chickpea, lentil, peas. *Rabi* redgram and, rajmash. Oilseed- rapeseed, mustard, sunflower, safflower; and linseed. Sugar crops- sugarcane and sugar beet. Medicinal and aromatic crops- mentha, lemon grass and citronella. Forage crops –barseem, lucerne and oat; potato, quinoa, tobacco.

Practical

Sowing methods of wheat and sugarcane; identification of weeds in *rabi* season crops; Working out seed rate and real value related numericals, study of morphological characteristics of *rabi* crops; study of yield contributing characters of *rabi* season crops; yield and juice quality analysis of sugarcane; study of important agronomic experiments of *rabi* crops at experimental farms; Top dressing and foliar feeding of nutrients, study of *rabi* forage experiments; oil extraction of medicinal crops; visit to research stations of related crops.

Lecture schedule: Theory

S.N.	Topic	No. of lectures
1.	Wheat- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
2.	Barley- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
3.	Chickpea-Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
4.	Lentil- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
5.	Rajmah -Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
6.	Rapeseed and mustard- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
7.	Sunflower- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
8.	Safflower - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
9.	Linseed - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
10.	Sugarcane- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
11.	Sugarbeet - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
12.	Mentha, Lemon grass and Citronella - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
13.	Barseem, Lucerne and Oat - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
14.	Potato - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1

15.	Quinoa - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
16.	Tobacco - Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1

Lecture schedule: Practical

S.N.	Topic	No. of lectures
1.	Sowing methods of wheat and sugarcane	3
2.	Identification of weeds in <i>rabi</i> season crops	3
3.	Study of morphological characteristics of <i>rabi</i> crops	3
4.	Working out seed rate and real value related numericals	2
5.	Study of yield contributing characters of <i>rabi</i> season crops	3
6.	Yield and juice quality analysis of sugarcane	3
7.	Study of important agronomic experiments of <i>rabi</i> crops at experimental farms	3
8.	Top dressing and foliar feeding of nutrients	3
9.	Study of <i>rabi</i> forage experiments	3
10.	Oil extraction of medicinal crops	3
11.	Visit to research stations of related crops	3

Suggested Readings

1. B. Gurarajan, R. Balasubramanian and V. Swaminathan. Recent Strategies on Crop Production. Kalyani Publishers, New Delhi.
2. Chidida Singh. 1997. Modern techniques of raising field crops. Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi.
3. Rajendra Prasad. Textbook of Field Crops Production - Commercial Crops. Volume II ICAR Publication.
4. Rajendra Prasad. Textbook of Field Crops Production - Foodgrain Crops. Volume I ICAR Publication.
5. S.R. Reddy. 2009. Agronomy of Field Crops. Kalyani Publishers, New Delhi.
6. S.S. Singh. 2005. Crop Management. Kalyani Publishers, New Delhi.
7. Rajendra Prasad. 2002. Text Book of Field Crops Production, ICAR, New Delhi.
8. Reddy, S.R. 2004. Agronomy of Field crops, Kalyani Publishers, Ludhiana.
9. Subhash Chandra Bose, M. and Balakrishnan, V. 2001. Forage Production South Asian Publishers, New Delhi.
10. UAS, Bangalore. 2011. Package of Practice. UAS, Bengaluru.

AGR0N-222	Water Management	2(1+1)
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Objectives

1. To study the fundamental concept about scientific management of irrigation water to achieve higher water productivity.
2. This course is aimed at acquainting the students about the fundamental principles of water management along with various irrigation practices at the field level.

Theory

Introduction - definitions, objectives of irrigation and water management, importance of irrigation water, terminologies used in water management and irrigation. History of irrigation in India. Water resources - distribution of water resources, rainfall, reservoirs and tanks, underground water resources, irrigated area in India and Karnataka under the tanks, reservoirs and wells and hydrological cycle. Exploitation of Water Resources - dams, canals, underground, lift irrigation and waste water use systems. Soil water relations - physical nature and characteristics of soil - adhesion, cohesion, capillarity, porosity, bulk density and particle density; effective root zone depth; nature of soil water - hygroscopic, capillary and gravitational water, soil moisture constants, infiltration, expression of soil moisture, soil moisture tension and potentials, expression of soil moisture and their mutual relations, hysteresis, hydro dynamics, assessment of soil moisture by direct methods - gravimetric, volumetric, rapid moisture meter and sulphuric acid method; Indirect methods – tensiometer, gypsum block/ resistance block meter and neutron moisture meter. Plant Water Relations - importance of water in plants, critical stages, effects of excess and deficit moisture on plant growth and water availability and nutrient uptake. Assessment of Water Requirement (WR) - components and components of water requirement- water supplied and water demand, factors influencing water requirement, methods to estimate water requirement - lysimeter, Estimation of ET losses, soil moisture depletion, field experimentation technique, water balance method, drum culture technique, expressions of water requirement. Methods of irrigation - classification, principles, suitability, advantages and disadvantages of surface irrigation methods - flooding, border strip, check basin, ridges and furrow and subsurface irrigation methods. Sprinkler irrigation method - types of sprinkler irrigation, components, advantages and disadvantages, water distribution pattern and wetting area based on wind speed. Drip irrigation method- differences between sprinkler and drip irrigation methods, types of drip irrigation, components, advantages and disadvantages, water distribution pattern, pitcher irrigation, cablegation, fertigation and suitability of irrigation methods. Irrigation efficiency - conveyance, application of water, storage of water in soil profile, distribution of water, Water Use Efficiency (WUE) - types and methods to improve WUE. Assessment of Irrigation Requirement (IR) - meaning, net irrigation requirement, gross irrigation requirement and relationship between IR and WR. Quantitative estimation of irrigation water - units of water measurement - direct methods - volume time relation, velocity area, tracer method, current meter, coordinate method. Indirect methods - weirs, orifices and parshall flume, duty of water. Suitability of land water for irrigation – Suitability of land – Physical assessment: texture, water holding capacity,

non-capillary pores and permeability, degree of slope, direction of slope, depth of water table. Chemical assessment: soluble salt content, exchangeable sodium, pH. Quality of irrigation water - total soluble salts, sodium adsorption ratio; effects of poor quality of water on crop performance, Irrigation management in problematic areas – poorly drained

area, saline and alkaline lands. Scheduling of irrigation - definition, approaches to scheduling - Feel and appearance method, plant / crop indices, indicator plants, critical stage approach, soil moisture depletion approach, IW / CPE ratio approach, stress degree day (SDD) concept. Irrigation control structure and water conveyance methods - unlined and lined open channels,

pipes, drop structures, check gates, chute spillways, turn outs, flumes and culverts. Irrigation practices for some important crops - paddy, closed and wide spaced crops, plantation and orchard crops and nurseries, suitable irrigation practice for different methods of mulberry cultivation. Drainage - surface and sub-surface methods.

Practical

Soil moisture determination by direct methods. Study and installation of tensiometer. Study and installation of gypsum block resistance moisture meter. Determination of bulk density. Determination of maximum water holding capacity. Determination of field capacity. Determination of permanent wilting point. Determination of infiltration rate. Study of capillarity in soil. Study of methods of water flow measurement. Study and use of weirs and orifices. Study and use of parshall flume. Study of surface and sub-surface irrigation methods. Study of sprinkler irrigation. Study of drip irrigation. Study of water requirement calculation for mulberry. Study of on-farm irrigation structures. Study of drainage structures. Practice of numerical examples and conversions.

Lecture schedule: Theory

S.N.	Topic	No. of lectures
1.	Introduction - definitions, objectives of irrigation and water management, importance of irrigation water, terminologies used in water management and irrigation. History of irrigation in India.	1
2.	Water resources - distribution of water resources, rainfall, reservoirs and tanks, underground water resources, irrigated area in India and Karnataka under the tanks, reservoirs and wells and hydrological cycle.	1
3.	Exploitation of Water Resources - dams, canals, underground, lift irrigation and waste water use systems.	1
4.	Soil water relations - physical nature and characteristics of soil - adhesion,	1

	cohesion, capillarity, porosity, bulk density and particle density; effective root zone depth; nature of soil water - hygroscopic, capillary and gravitational water, soil moisture constants, infiltration, expression of soil moisture, soil moisture tension and potentials, expression of soil moisture and their mutual relations, hysteresis, hydro dynamics, assessment of soil moisture by direct methods - gravimetric, volumetric, rapid moisture meter and sulphuric acid method; Indirect methods – tensiometer, gypsum block/ resistance block meter and neutron moisture meter.	
5.	Plant Water Relations - importance of water in plants, critical stages, effects of excess and deficit moisture on plant growth and water availability and nutrient uptake.	1
6.	Assessment of Water Requirement (WR) - components and components of water requirement - water supplied and water demand, factors influencing water requirement, methods to estimate water requirement - lysimeter, Estimation of ET losses, soil moisture depletion, field experimentation technique, water balance method, drum culture technique, expressions of water requirement.	1
7.	Methods of irrigation - classification, principles, suitability, advantages and disadvantages of surface irrigation methods - flooding, border strip, check basin, ridges and furrow and subsurface irrigation methods.	1
8.	Sprinkler irrigation method - types of sprinkler irrigation, components, advantages and disadvantages, water distribution pattern and wetting area based on wind speed.	1
9.	Drip irrigation method- differences between sprinkler and drip irrigation methods, types of drip irrigation, components, advantages and disadvantages, water distribution pattern, pitcher irrigation, cablegation, fertigation and suitability of irrigation methods.	1
10.	Irrigation efficiency - conveyance, application of water, storage of water in soil profile, distribution of water, Water Use Efficiency (WUE) - types and methods to improve WUE.	1
11.	Assessment of Irrigation Requirement (IR) - meaning, net irrigation requirement, gross irrigation requirement and relationship between IR and WR.	1
12.	Quantitative estimation of irrigation water - units of water measurement -	1

	direct methods - volume time relation, velocity area, tracer method, current meter, coordinate method. Indirect methods - weirs, orifices and parshall flume, duty of water.	
13.	Suitability of land water for irrigation – Suitability of land – Physical assessment: texture, water holding capacity, non-capillary pores and permeability, degree of slope, direction of slope, depth of water table. Chemical assessment: soluble salt content, exchangeable sodium, pH. Quality of irrigation water - total soluble salts, sodium adsorption ratio; effects of poor quality of water on crop performance, Irrigation management in problematic areas – poorly drained area, saline and alkaline lands.	1
14.	Scheduling of irrigation - definition, approaches to scheduling - Feel and appearance method, plant/crop indices, indicator plants, critical stage approach, soil moisture depletion approach, IW / CPE ratio approach, stress degree day (SDD) concept.	1
15.	Irrigation control structure and water conveyance methods - unlined and lined open channels, pipes, drop structures, check gates, chute spillways, turn outs, flumes and culverts.	1
16.	Irrigation practices for some important crops - paddy, closed and wide spaced crops, plantation and orchard crops and nurseries, suitable irrigation practice for different methods of mulberry cultivation. Drainage - surface and sub-surface methods.	1

Lecture schedule: Practical

S.N.	Topic	No. of lectures
1	Soil moisture determination by direct methods	1
2	Study and installation of tensiometer	1
3	Study and installation of gypsum block resistance moisture meter	1
4	Determination of bulk density	1
5	Determination of maximum water holding capacity	1
6	Determination of field capacity and permanent wilting point	1
7	Determination of infiltration rate	1
8	Study of capillarity in soil	1
9	Study of methods of water flow measurement	1

10	Study and use of weirs, orifices and parshall flume	1
11	Study of surface and sub-surface irrigation methods	1
12	Study of sprinkler irrigation & drip irrigation	1
13	Study of water requirement calculation for mulberry	1
14	Study of on-farm irrigation structures	1
15	Study of drainage structures	1
16	Practice of numerical examples and conversions	1

Suggested readings

1. Irrigation and Water Management - Mukund Joshi and Prabhakar Setty T.K. 2013, Kalyani Publishers.
2. Irrigation Agronomy - Reddy, S.R. 2016, Kalyani Publishers.
3. Irrigation Management - Darra, B.L. and Raghuvanshi, C.S. 1990, M/S Atlantic Publishers and Distributors.
4. Irrigation Water Management: Principles and Practices - Dilipkumar Mujumdar, 2014, Prentice Hall India Learning Private Limited.
5. Irrigation: Theory and Practice - Michael, A.M., 2007, Vikas Publishing House Pvt. Limited.
6. Agricultural Water Management - Premjit Sharma, 2008, Daya Publishing House.

AENGG-221	Farm Machinery and Power	2(1+1)
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Objectives

To enable the students to understand the need of farm power, basic principles and parts of IC engine, different tillage, sowing, intercultural, plant protection equipment, working principles of threshers, harvesting of field and horticultural crops.

Theory

Status of Farm Power in India; Sources of Farm Power, I.C. engines, working principles of I C engines; comparison of two stroke and four stroke cycle engines, Study of different components of I.C. engine, I.C. engine terminology and solved problems; Familiarization with different systems of I.C. engines: Air cleaning, cooling, lubrication, fuel supply and hydraulic control system of a tractor; Familiarization with Power transmission system: clutch; gear box, differential and final drive of a tractor; Tractor types; Cost analysis of tractor power and attached implement; Criteria for selection of tractor and machine implements. Familiarization with Primary and Secondary Tillage implement; Implement for hill agriculture; implement for intercultural operations; Familiarization with sowing and planting equipment; calibration of a seed drill and solved examples; Familiarization with Plant Protection equipment; Familiarization with harvesting and threshing equipment.

Practical

Study of different components of I.C. engine. To study air cleaning and cooling system of engine; Familiarization with clutch, transmission, differential and final drive of a tractor; Familiarization with lubrication and fuel supply system of engine; Familiarization with brake, steering, hydraulic control system of engine; Learning of tractor driving; Familiarization with operation of power tiller; Implements for hill agriculture; Familiarization with different types of primary and secondary tillage implements: mould plough, disc plough and disc harrow; Familiarization with seed-cum-fertilizer drills their seed metering mechanism and calibration, planters and trans planter; Familiarization with different types of sprayers and dusters; Familiarization with different inter-cultivation equipment; Familiarization with harvesting and threshing machinery; Calculation of power requirement for different implements.

Lecture schedule : Theory

S.No.	Topic	No. of lectures
1.	Sources of farm power and its status in India and Rajasthan.	1
2.	I.C. engines, working principles of I C engines, comparison of two stroke and four stroke cycle engines	1
3.	Study of different components of I.C. engine, I.C. engine terminology and numerical.	2
4.	Air supply and exhaust system Pre cleaners, oil soaked element oil bath type air cleaners; Fuel supply system	1
5.	Lubricating system- splash system and forced feed system; Cooling system- thermosiphon system and forced circulation system	1
6.	Transmission system- clutch, gear box, differential, final drive, P.T.O. and hydraulic control system	1
7.	Tractor types, Cost analysis of tractor power and attached	1
8.	Familiarization with primary and secondary tillage implement	2
9.	Criteria for selection of tractor and machine implements.	1
10.	Familiarization with implement for intercultural operations	1
11.	Familiarization with sowing and planting equipment,	1
12.	Familiarization with Plant Protection equipment	1
13.	Familiarization with harvesting and threshing equipment	2

Lecture schedule: Practical

S.No.	Topic	No. of lectures
1	Study of different components of I.C. engine.	1
2	To study air cleaning and fuel supply system of engine,	1
3	Study of cooling and lubricating system.	1
4	Study of transmission system clutch, gear box, differential, final drive and P.T.O.	1
5	Familiarization with brake, steering, hydraulic control system of	1

6	Tractor driving	2
7	Daily and periodic maintenance of tractor	1
8	Study of power tiller and garden tractor	1
9	Study of primary and secondary tillage implements: mould board disc plough	1
10	Study of secondary tillage implements-	1
11	Study of seed metering mechanism and calibration of seed d numerical	2
12	Study of different types of sprayers and dusters	1
13	Study of reaper and thresher	2

Suggested readings:

1. Jagdiswar Sahay – Elements of Agricultural Engineering, Agro. Book Agency, New Chitragupta Nagar, Patna
2. Jain, S.C. and C.R. Rai-Farm Tractor and maintenance and repair. Standard Publishers, 1705-B, Naisarak. Delhi- 110006
3. Ojha, T.P. and A.M. Michael, A.M. Principles of Agricultural Engineering. Vol.I. Jain brothers, 16/893, East Park Road, Karol Bagh, New Delhi -110005
4. Surendra Singh- Farm machinery –Principles and applications, ICAR, New Delhi
5. Farm Tractors, Maintenance and Repair.1989. Rai and Jain. Tata Mc Graw Hill Publ. New Delhi.
6. Elements of Farm Machinery.1989. Srivastava, A.C. Oxford IBH Publ. Company, New Delhi.

AGEXT-221	Entrepreneurship Development and Business Communication	3(2+1)
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Objective

1. To provide student an insight into the concept and scope of entrepreneurship
2. To expose the student to various aspects of establishment and management of a small business unit
3. To enable the student to develop financially viable agribusiness proposal

Theory

Development of entrepreneurship, motivational factors, social factors, environmental factors, characteristics of entrepreneurs, entrepreneurial attributes/competencies. Concept, need for and importance of entrepreneurial development. Evolution of entrepreneurship, objectives of entrepreneurial activities, types of entrepreneurs, functions of entrepreneurs, importance of entrepreneurial development, and process of entrepreneurship development. Environment scanning and opportunity identification need for scanning: spotting of opportunity, scanning of environment identification of product / service: starting a project; factors influencing sensing the opportunities. Infrastructure and support systems: good policies, schemes for entrepreneurship development; role of financial institutions, and other agencies in entrepreneurship development. Steps involved in functioning of an enterprise. Selection of the product / services, selection of form of ownership; registration, selection of site, capital

sources, acquisition of manufacturing know how, packaging and distribution. Planning of an enterprise, project identification, selection, and formulation of project; project report preparation, Enterprise Management. Production management: product, levels of products, product mix, quality control, cost of production, production controls, Material management. Production management: raw material costing, inventory control. Personal management: manpower planning, labour turn over, wages / salaries. Financial management /accounting: funds, fixed capital and working capital, costing and pricing, long term planning and short-term planning, book keeping, journal, ledger, subsidiary books, annual financial statement, taxation. Marketing management: market, types, marketing assistance, market strategies. Crisis management: raw material, production, leadership, market, finance, natural etc.

Practical

Visit to small scale industries/agro-industries, Interaction with successful entrepreneurs/agricentrepeneurs. Visit to financial institutions and support agencies. Preparation of project proposal for funding by different agencies.

Suggested Readings

1. Charantimath, P.M. 2009, Entrepreneurship Development and Small Business Enterprises. Pearson Publications, New Delhi.
2. Desai, V. 2015, Entrepreneurship: Development and Management, Himalaya Publishing House.
3. Gupta, C.B. 2001. Management Theory and Practice. Sultan Chand & Sons.
4. Indu Grover. 2008. Handbook on Empowerment and Entrepreneurship. Agrotech Public Academy.
5. Khanka, S.S. 1999. Entrepreneurial Development. S. Chand & Co.
6. Mehra, P. 2016, Business Communication for Managers. Pearson India, New Delhi.
7. Pandey, M. and Tewari, D. 2010, The Agribusiness Book. IBDC Publishers, Lucknow.
8. Singh, D. 1995. Effective Managerial Leadership. Deep & Deep Publ.
9. Singhal, R.K. 2013, Entrepreneurship Development & Management, Katson Books.
10. Tripathi, P.C. and Reddy, P.N. 1991. Principles of Management. Tata McGraw Hill.
11. Vasant Desai, 1997. Small Scale Industries and Entrepreneurship. Himalaya Publ. House.

GPB-221	Basics of Plant Breeding	3(2+1)
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Objectives

To acquaint with different techniques ranging from simply selecting plants with desirable characteristics for propagation, to more complex molecular techniques for breeding new varieties, which are higher yielding, resistant to biotic and abiotic stresses for ensuring food security.

Theory

Historical development, concept, nature and role of plant breeding, major achievements and future prospects; Genetics in relation to plant breeding, modes of reproduction and apomixes, self-incompatibility and male-sterility-genetic consequences, cultivar options, Plant genetic resources, its utilization and conservation Domestication, Acclimatization and Introduction. Centres of origin/

diversity, Components of Genetic variation. Heritability and genetic advance. Pre-breeding and Universal Plant Breeder's equation. Genetic basis and breeding methods in self-pollinated crops- mass and pure line selection, hybridization techniques and handling of segregating population. Multiline concept, Concepts of population genetics and Hardy-Weinberg Law, Genetic basis and methods of breeding cross-pollinated crops, modes of selection. Population movement schemes- Ear to Row method, Modified Ear to Row, recurrent selection schemes. Heterosis and inbreeding depression, development of inbred lines and hybrids, composite and synthetic varieties. Breeding methods in asexually propagated crops, clonal selection and hybridization. Wide hybridization and pre-breeding. Polyploidy in relation to plant breeding, mutation breeding- methods and uses. Breeding for important biotic and abiotic stresses. Participatory plant breeding. Variety Release and notification. Intellectual Property Rights, Patenting, Plant Breeders and Farmer's Rights.

Practical

Plant Breeder's kit, Study of germplasm of various crops, Study of floral structures of self- pollinated and cross-pollinated crops, Emasculation and hybridization techniques in self and cross pollinated crops, Consequences of inbreeding on genetic structure of resulting populations, Study of male sterility system, Handling of segregating populations, Methods of calculating mean, range, variance, standard deviation, heritability, Designs used in plant breeding experiments, analysis of Randomized Block Design, To work out the mode of pollination in a given crop and extent of natural out-crossing, Prediction of performance of double cross hybrids, Maintenance of breeding records and data collection, Screening tests for biotic and abiotic stresses.

Lecture Schedule: Theory

S. No.	Name of topic	No. of Lectures
1	Plant breeding: concept, nature, objectives and role of plant breeding	1
2	Historical development of plant breeding	1
3	Major achievements and future prospects	1
4	Genetics in relation to plant breeding	1
5	Modes of reproduction and pollination, apomixes	1
6	Self – incompatibility	1
7	Male sterility- genetic consequences	1
8	Domestication, Acclimatization, introduction, Centre of origin/diversity	1
9	Component of Genetic variation; Heritability and genetic advance	1
10	Genetic basis of self- pollinated crops and pure line theory	1
11	Breeding methods in self- pollinated crops- mass and pure line selection	1

12	Hybridization techniques	1
13	Handling of segregating population (pedigree, bulk and back cross method)	1
14	Multiline concept	1
15	Concepts of population genetics and Hardy-Weinberg Law	1
16	Genetic basis and methods of breeding cross pollinated crops	1
17	Population improvement and modes of selection	1
18	Heterosis and inbreeding depression	1
19	Development of inbred lines and hybrids	1
20	Composite and synthetic varieties	1
21	Breeding methods in asexually propagated crops	1
22	Clonal selection and hybridization	1
23	Wide hybridization and pre-breeding	1
24	Polyploidy in relation to plant breeding	1
25	Mutation breeding- methods and uses	1
26	Breeding for important biotic and abiotic stresses	1
27	Breeding for important abiotic stresses	1
28	Biotechnological tools-DNA markers	1
29	Marker assisted selection	1
30	Participatory plant breeding	1
31	Intellectual Property Rights and Patenting	1
32	Plant Breeders and & Farmer's Rights	1

Lecture Schedule: Practical

S.N.	Topic	No. of lectures
1	Plant Breeder's kit	1
2	Study of germplasm of various crops	1
3	Study of floral structure of self pollinated crops	1
4	Study of floral structure of cross pollinated crops	1
5	Emasculation and hybridization techniques in self pollinated crops I	1
6	Emasculation and hybridization techniques in self pollinated crops II	1
7	Emasculation and hybridization techniques in self & cross pollinated crops	1
8	Emasculation and hybridization techniques in self & cross pollinated crops	1
9	Consequences of inbreeding on genetic structure of resulting populations	1
10	Study of male sterility system	1
11	Handling of segregating populations	1
12	Methods of calculating mean, range, variance, standard deviation, heritability	1
13	Designs used in plant breeding experiment	1
14	Analysis of Randomized Block Design and components of genetic variance	1
15	To work out the mode of pollination in a given crop and extent of natural out crossing	1

Suggested Readings

1. Principles of Plant Breeding (1st & 2nd Edition) by RW Allard.
2. Plant Breeding: Principles & Practices by JR Sharma.
3. Plant Breeding- B.D. Singh.
4. Principles and Procedures of Plant Breeding - Biotechnical and Conventional Approaches by GS Chahal and SS Gosal.
5. Principles of Plant Genetics and Breeding by George Acquaah.

HORT-221	Production Technology for Vegetables and Spices	2(1+1)
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Objectives

1. To educate about the different forms of classification of vegetables
2. To educate about the origin, area, climate, soil, improved varieties and cultivation practices of vegetables and spices.
3. To educate about the physiological disorders of vegetables and spices

Theory

Importance of vegetables and spices in human nutrition and national economy, kitchen gardening, brief about origin, area, climate, soil, improved varieties and cultivation practices such as time of sowing, transplanting techniques, planting distance, fertilizer requirements, irrigation, weed management, harvesting and yield, physiological disorders of important vegetable and spices (tomato, okra, brinjal, chili, capsicum, cucumber, bitter gourd, bottle gourd, sweet potato, cassava and moringa, pumpkin, French bean, peas; cole crops such as cabbage, cauliflower, knol-khol; bulb crops such as onion, garlic; root crops such as carrot, radish, beetroot; tuber crops such as potato; leafy vegetables such as amaranth, palak, perennial vegetables, spice crops like turmeric, zinger, garlic, coriander, cumin, black pepper, cardamom, fenugreek, fennel, clove, nutmeg, cinnamon, curry leaf, tamarind and herbal spices).

Practical

Identification of vegetables and spice crops and their seeds. Description of varieties. Propagation methods - rapid multiplication techniques - seed collection and extraction. Nursery raising. Direct seed sowing and transplanting. Study of morphological characters of different vegetables and spices. Fertilizers applications. Harvesting and post-harvest practices, Economics of vegetables and spices cultivation, visit to spice gardens.

Lecture schedule: Theory

S. No.	Topics	No. of lectures
1.	Importance of vegetables and spices in human nutrition and national economy	1
2.	Kitchen gardening and types of vegetable gardens	1
3.	Brief about origin, area, climate, soil for vegetable crops, improved varieties and cultivation practices of Solanaceous crops such as tomato Brinjal, chilli and capsicum	2
4.	Cucurbitaceous crops (Cucumber, Watermelon, Muskmelon, Bitter gourd, Bottle gourd, Pumpkin)	1
5.	Leguminous crops (Pea, French bean)	1

6.	Tuber crops (Potato, Sweet potato, Cassava)	1
7.	Cole crops (Cabbage, Cauliflower, Knol-khol)	1
8.	Bulb crops (Onion and Garlic)	1
9.	Root crops (Carrot, Radish, Beetroot)	1
10.	Leafy vegetable such as Amaranth, Palak	1
11.	Perennial vegetables - Moringa	1
12.	Spices crops like black pepper and cardamom	1
13.	Turmeric and ginger	1
14.	clove, nutmeg, cinnamon, curry leaf, tamarind and herbal spices	1
15.	Seed spices such as cumin, fenugreek, fennel, coriander	1
		16

Lecture schedule: Practical

S. No.	Topics	No. of lectures
1.	Identification of vegetables and spice crops and their seeds.	2
2.	Description of vegetable and spices crop varieties.	2
3.	Propagation methods - rapid multiplication techniques.	2
4.	Seed collection and extraction.	1
5.	Nursery raising.	1
6.	Direct seed sowing and transplanting.	1
7.	Study of morphological characters of different vegetables crops.	1
8.	Study of morphological characters of different spices crops.	1
9.	Fertilizers applications in vegetable and spices crops.	1
10.	Harvesting and post-harvest practices of vegetable and spices crops.	2
11.	Economics of vegetables and spices cultivation.	1
12.	Visit to commercial vegetables /spice gardens.	1
		16

Suggested readings

1. Olericulture, Fundamentals of Vegetable Production (Vol.1) by K.P. Singh, Anant Bahadur
2. Vegetable crops by J. Kabir, T.K. Bose, M.G. Som
3. Vegetable crops (Production technology, Vol II) by M.S. Fagaria, B.R. Choudhury, R.S. Dhaka
4. Bosse, T. K. 2010. Vegetable Crops. Naya Prakash, Calcutta.
5. Chadha, K. L. 2003. Handbook of Horticulture. ICAR, New Delhi.
6. Choudhary, B. R. 2009. A Text book on production technology of vegetables. Kalyani Publishers.
7. Dhaliwal, M. S. 2008. Handbook of Vegetable Crops. Kalyani Publishers.

NCC-IV/ NSS-IV/PHED-IV	NSS/NCC/Physical Education & Yoga Practices**	1(0+1)
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NCC-IV	National Cadet Corps**	1(0+1)
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1. The grid system. Relief, contours and gradients. Cardinal points and finding north. Types of bearings and use of service protractor.
2. Prismatic compass and its use. Setting a map, finding north and own position. Map to ground and ground to map.
3. Knots and lashings, Camouflage and concealment, Explosives and IEDs.
4. Field defenses obstacles, mines and mine lying. Bridging, waterman ship
5. Field water supplies, tracks and their construction.
6. Nuclear, Chemical and Biological Warfare (NCBW)
7. Judging distance. Description of ground and indication of landmarks.
8. Recognition and description of target. Observation and concealment. Field signals. Section formations.
9. Fire control orders. Fire and movement. Movement with/without arms. Section battle drill.
10. Types of communication, media, latest trends and developments.

NSS-IV	National Service Scheme **	1(0+1)
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1. **Youth and crime:** Sociological and psychological factors influencing youth crime, cyber crime, peer mentoring in preventing crime and awareness for juvenile justice
2. **Civil/self defence:** Civil defence services, aims and objectives of civil defence; needs and training of self defence Resource mobilisation
3. Writing a project proposal of self fund units (SFUs) and its establishment Additional life skills
4. Positive thinking, self confidence and esteem, setting life goals and working to achieve them, management of stress including time management.

PHED-IV	Physical Education & Yoga Practices**	1(0+1)
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1. Teaching of different field events—demonstration practice of the skills and correction.
2. Teaching of different field events— demonstration practice of the skills and correction with competition among them.
3. Teaching of different asanas—demonstration practice and correction.
4. Teaching of weight training—demonstration practice and correction.

5. Teaching of circuit training—demonstration practice and correction.
6. Teaching of calisthenics—demonstration practice and correction.